# Predicting Tool Extension in VR Interactions Using Gaze Features

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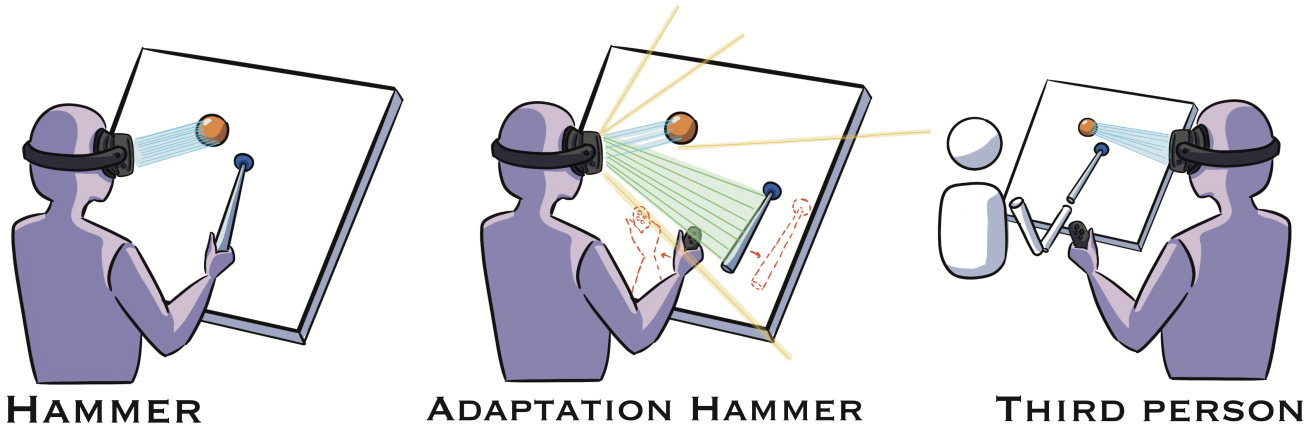



**Figure 1: Overview of experimental conditions and hypothesized gaze behavior across tool extension states. Participants perform a button-pressing task in VR using three tool configurations: a standard hammer (left), an adaptation hammer with altered control mappings (middle), and a third person avatar-controlled hammer (right). Tool extension is characterized by task-directed gaze, while breakdowns in extension lead to increased attention to the tool and surrounding environment.**

## Abstract

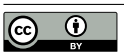
The ability for virtual reality (VR) systems to create a sense of presence in users has long captured the attention of HCI researchers. One aspect of presence during interaction is tool extension - the experience of integrating a tool into one's body schema. Prior research shows that features of visual attention correlate with experiencing tool extension in VR, but no previous methods provide an ability to predict a user's tool extension state in real-time. We investigate building this predictor through a study with 18 participants. Participants used virtual tools designed to either promote or inhibit tool extension (a simple hammer vs. an unpredictable linkage-driven hammer). VR gaze-tracking data reliably distinguished these conditions based on the distribution of gaze across areas of interest such as tool, task, and elsewhere. We also found that perspective (first vs. third person) did not affect tool extension. Our results suggest a straightforward gaze-based predictor for tool extension. We conclude by providing a design space demonstrating the practical value of this predictor and several applications enabled by it.

## CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*; *Interaction techniques.*

## Keywords

Virtual Reality, Eye Tracking, Gaze Analysis, Tool Extension, Embodied Interaction



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## 1 Introduction

Humans extend their capabilities through tools. This isn't just a functional difference, it changes how we perceive and act within the world. Researchers describe this phenomenon as integrating the tool with our bodies, updating our body schema [8, 33]. Classic examples of this range from using objects in the physical world to computer interfaces for digital systems, such as how a blind person using a cane does not perceive the cane [35], but instead perceives the world through the tip, to the understanding of how humans can type documents without thinking about pressing keys [52]. This is also referred to as tool extension, and it is not a binary property. It is a spectrum from deliberate, effortful control to seamless, transparent use.

One framework for reasoning about tool extended states comes from Phenomenology. As described by Winograd and Flores in *Understanding Computers and Cognition* [52]: a **ready-to-hand** state occurs when the tool disappears from conscious attention, and the user acts through it while being focused on the task, whereas a **present-at-hand** state occurs when the tool itself is the object of attention as users struggle with or learn to use it.

In the field of Psychology, eye tracking has been used extensively to study how eye gaze supports manual task performance. In a study of eye movements during manual tasks, it was found that the eyes exhibit different patterns at different task stages [28]. A study of eye tracking across several professional domains [12] showed that the eye behaves differently as expertise is acquired. Experts tend to focus on task-relevant areas while novices tend to distribute their attention more broadly. Gaze has also been linked to flow states, the psychological state of being absorbed in an action [14]. These findings establish that there is a relationship between a user's eyes and their cognitive relationship with their task and tool.

Prior work in HCI has explored the connection of computer tool use to these states [3, 6], providing a lens that tool extension can occur in VR. One insight from this work is that attention to task vs attention to tool maps onto ready-to-hand and present-at-hand states. However, these studies often rely on users self-reporting attention shifts, or carefully designing studies to demonstrate the concept exists. The missing piece is a continuous sensing modality that captures these attention shifts as they happen.

If we could detect the user's tool integration state as it changes moment-to-moment, it enables new design possibilities for things such as adaptive assistance, skill training, and tool evaluation. Currently, we rely on subjective reports or task performance as proxies. These are post-hoc and coarse, and they don't easily capture the shifts between ready-to-hand and present-at-hand that can happen in an open-ended task. This motivates the development of real-time, objective measures of tool integration states. We hypothesize that real-time gaze tracking can be used to predict the user's tool extension state.

We situate our exploration in Virtual Reality for two reasons. First, VR provides an especially rich space for exploring tool extension, as it allows for tool relationships that are physically impossible. You can give users a stretched arm [43] or even multiple bodies [36, 46]. The design space for VR tools is not constrained by our physical morphology, and this further complicates the picture of how users integrate with tools that have no physical analog. Second, modern VR headsets such as the Meta Quest Pro and Apple Vision Pro now include built-in eye tracking technology. With eye tracking no longer constrained to research devices, it is available as a standard input channel for interaction. This opens up the possibility of real-time, always-on assessment of a user's tool extension state. Our research question is the following: **can we use this readily available gaze data to detect whether a user is in a ready-to-hand or present-at-hand state with their virtual tool?**

To explore this question, we run a study analyzing various tool extension relationships. We give the users an everyday, easy-to-understand task to be completed using three different tools spanning the spectrum of tool integration. First, a regular hammer, which represents skilled natural tool use (ready-to-hand is expected). Second, a hammer which requires constant sensorimotor adaptation, disrupting the mapping between hand movement and tool movement (expected to induce present-at-hand state). Third, using a hammer from a third person perspective, a tool relationship unique to the medium of VR, and whose tool extension properties are unknown. We analyze gaze allocation patterns across conditions and relate them to the ready-to-hand and present-at-hand framework.

We also identify specific gaze features (tool gaze, elsewhere gaze) that differentiate tool extension from tool breakdown states. These features are used to develop and validate a real-time classifier that predicts user difficulty during open-ended tool usage.

Our contributions are as follows:

- A study exploring the relationship between gaze patterns and tool extension states
- An analysis of which gaze features differentiate breakdown from tool extension
- The development of a real-time classifier that predicts current tool extension states
- A design space and sample applications of the usefulness of the real-time predictive metric

## 2 Related Work

Our work draws on research in tool integration, eye tracking during skilled action, and gaze-based machine learning classification to detect the transition between struggling and fluent tool use in VR.

### 2.1 Tool Integration

The phenomenological concepts of ready-to-hand and present-at-hand have been taken up across multiple disciplines to study how humans relate to their tools.

In neuroscience, studies have shown that tool use physically alters the brain's representation of the body, neurons in the parietal cortex extend their receptive fields to incorporate a held tool, and reaching behavior changes after tool use as if the body schema has been updated to include the tool [8, 33]. These works establish that tool integration is not just a philosophical concept, but a measurable neural phenomenon.

Winograd and Flores brought these concepts to computing with Understanding Computers and Cognition [52], arguing that effective computer use is characterized by ready-to-handness (the system is transparent and the user focuses on their work), while

errors or unfamiliar interfaces cause a breakdown, shifting the system to present-at-hand. Dourish further developed these ideas for HCI in *Where the Action Is* [10], arguing that meaning in interactive systems arises from skillful, embodied engagement rather than mental representation, and that the transparency of well-designed tools during skilled use is a central design goal.

Our work is heavily inspired by two sets of work from the HCI literature. Bergström et al. introduced a quantitative measure of tool extension in HCI using a visual-tactile interference paradigm, demonstrating that both traditional input devices and VR controllers produce measurable integration into the user's body schema [6]. They found that realistic VR hand representations yield greater tool extension than abstract pointers, suggesting that the design of the virtual tool affects the degree of integration.

Alzayat et al. [3, 4] introduced the Locus-of-Attention Index (LAI), a quantitative metric that captures the proportion of attention directed toward the task versus the tool. They found that direct hand interaction produced higher embodiment than VR controllers, and that the LAI successfully distinguished between working and broken tools, validating it as a measure of ready-to-hand versus present-at-hand states.

One important distinction between the prior studies and our work is that they measured attention indirectly through things like reaction times or reactions to secondary stimuli. Our work takes a different approach by directly observing the eyes to make tool extension observations.

## 2.2 Eye Tracking in VR and Skill Assessment

There has been a significant amount of prior research into eye gaze based interactions in HCI, such as using the gaze to assist selection tasks [27, 44], using the gaze point as an anchor or frame of reference for other interactions [32, 40, 41, 41], and creating novel interactions between users and their environment [17, 29]. In these projects, the eye takes an active role in interaction, requiring explicit eye gaze behaviors to be used.

At the same time, there have been a number of projects that explore implicit uses of the eye [50], such as adapting the interface based on where the user is looking [34, 39, 48, 49], but these interfaces often treat the eye as a lone signal. Our work combines implicit gaze with the context of skilled tool use, using gaze patterns not as a standalone input but as a window into the user's relationship with their tool.

Eye movements during manual tasks are tightly coupled to action plans: gaze anticipates each manipulation by roughly half a second, fixates task-relevant objects almost exclusively, and rarely lands on the hands themselves [15, 28]. A meta-analysis across professional domains confirms that experts fixate task-relevant areas more, with shorter durations and longer saccades, while novices distribute attention less efficiently [12]. The quiet eye period (the final sustained fixation before a motor action) has been linked to flow states and skilled performance, with longer quiet eye durations mediating the relationship between attentional control and task outcomes [14]. Krejtz et al. [26] formalize this intuition with gaze transition entropy, a metric that captures how predictably attention shifts between areas of interest, providing a principled way to quantify the structure of scanning behavior.

These findings have increasingly been applied in VR. Broad reviews of eye tracking in head-mounted displays [1, 42] identify skill assessment and cognitive state inference as active but underexplored application areas, noting that most work remains in controlled laboratory settings. Hirzle et al. [18] contribute a design space for gaze interaction on HMDs that highlights the unique requirements of 3D gaze estimation. David-John et al. [9] demonstrate that gaze dynamics alone such as fixation detection, velocity, and the ambient-to-focal attention ratio can predict the intent to interact with VR objects at above-chance accuracy without any knowledge of scene content, establishing that internal state transitions are readable from eye movements in VR.

Our work weaves these threads together: we use gaze transition patterns between task-relevant AOIs such as the tool, the task target, and the surrounding environment as a direct, continuous signal for detecting whether a user has achieved transparent tool use in VR.

## 2.3 Gaze-Based Classifiers and Adaptive Systems

A growing body of work applies machine learning to eye tracking features for improved understanding of user intent [2, 24, 31, 37, 53] and attention [21, 47]. Shojaeizadeh et al. [45] use a Random Forest on 30 gaze features to detect task demand, while Kosch et al. [25] show that smooth pursuit degradation under cognitive load can be classified by an SVM across three difficulty levels. Vortmann et al. [51] compare classical feature-based approaches against encoding raw gaze time series as images for CNN classification of attentional states, finding that the image-based approach generalizes better across participants. In VR specifically, Hou et al. [20] classify head-gaze versus head-gesture movements using XGBoost on windowed kinematic features with leave-one-participant-out cross-validation.

More recent work has moved toward real-time prediction in applied settings. Nasri et al. [38] classify cognitive load during a complex VR assembly task using only fixation duration and pupil dilation with a multilayer perceptron. Kennel et al. [23] predict specific types of student difficulty from AOI-based gaze features, showing that classification improves monotonically as more of the task unfolds, demonstrating the viability of early, forward-looking detection. Langner et al. [30] provide a cautionary complement: in naturalistic field settings, gaze-based cognitive state detection fails to generalize across tasks, underscoring the importance of task-specific models. Our classifier follows this principle, training and evaluating within a single VR tool-use context using leave-one-participant-out cross-validation.

Unlike prior gaze classifiers that target cognitive load or attentional state, ours detects a phenomenological transition from present-at-hand to ready-to-hand using a forward-looking labeling strategy that enables prediction before the outcome is observable.

# 3 Experiment

The purpose of our experiment is to determine which eye gaze features are correlated with tool extension states while in VR. We give the user three tools to investigate this: one which is known to invoke tool extension, one which we designed to make tool extension difficult, and one which is unknown. We give the user

a simple repetitive button pressing task to invoke a tool extended state in the user.

Prior work suggests different tool extension states change the user's locus of attention [3] between different areas of interest: attention to action near the task vs attention to action near the tool. In this study we explore if this relationship holds by analyzing eye tracking features of the user's gaze patterns alone.

## 3.1 Study Design

The experiment followed a single factor within subjects design with three conditions (**tool type**).

The first tool condition we call hammer. It is attached to the user's hand and extends 0.6m in length, with a blue ball at the end that acts as an end effector for interacting with elements in the study. Prior theories suggest hand-held physical tools afford tool extension, so we use hammer as a baseline condition that can provide a similar interaction.

The adaptation-hammer tool forces the user to re-learn a sensorimotor adaptation between their hand and tool movement every time they press a button in the study task (see below). The purpose of this design is to have a condition where the user must constantly re-learn how to use the tool, making it very difficult for them to become tool extended.

Sensorimotor adaptations were forced in two ways. First, the movement axis for the XZ plane (forward/back and left/right) were rotated to a random starting direction. This would cause forward motions to move the controller to the left or right. This was uniformly sampled and randomized over a range from +60 to -60 degrees. Second, the X and Z axes were randomly flipped in addition to the rotation to induce further difficulty.

We also investigate how perspective affects tool extension through the 3p-hammer condition, where the user controls a simple hammer-holding avatar in the third person. In effect, the third person body is also a sensorimotor remapping, but one that provides an avatar as a visual anchor. It is currently unknown if a person can tool-extend into a third person body in VR.

Conditions were counterbalanced using all 6 combinations of ordering.

## 3.2 Task

The experimental task was to tap the end of the tool on buttons represented as orange spheres on a 1m x 1m board. This task resembles a game of whack-a-mole, where participants use a tool to tap on objects as they appear. This task was chosen due to its resemblance to the simplest interaction between a hammer-like tool and the environment. We also decided most participants would likely have some prior experience doing this activity in some form. A screenshot of the environment can be seen in Figure 2.

30 button positions were uniformly sampled across the board's surface, and their ordering was randomized to normalize the distance between any two sequential buttons in the list. This was done to minimize randomness around moving the tool between two very nearby or very far apart buttons.

Users were required to perform a tapping motion on the buttons. To achieve this, the tool's end effector would be clamped to the table's surface, preventing users from pushing the tool through the table. If the end effector was in a clamped state, the button cannot be pressed. This was implemented to prevent them from positioning the tool at the table's surface and dragging the tool to press buttons.

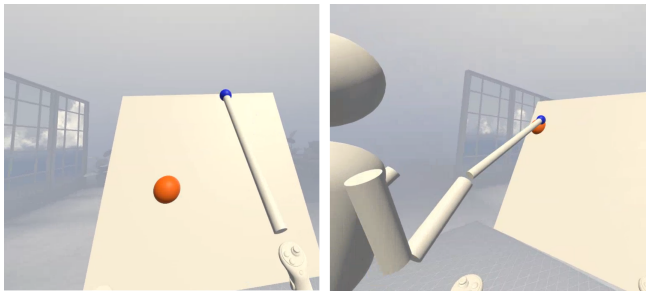


**Figure 2: Screenshots from our user study. (Left) Hammer condition (this also resembles the adaptation-hammer condition, but with controls remapped), (Right) 3p-hammer condition. The user is instructed to tap on the orange ball (task button). The end of the user's tool is blue, and it is the end effector of the tool. The rest of the environment is rendered in neutral colors.**

Users performed 30 button presses per experimental condition, and each set of conditions was run two times (e.g. A B C A B C). In between each condition, users could rest for as long as they wanted, and they could press a button on the VR controller to continue at their own pace. Once a condition started, users were instructed to press all 30 buttons before resting.

## 3.3 Measures

We measure a number of things as dependent variables.

First, we measure **task completion time**: how long each button press takes. All tools function the same (but behave differently), and so the effect of interface on mean task completion time can provide a measure of how "difficult" it was to use a particular tool.

The headset we used only provides a single combined eye ray, so individual eye positions and orientations were unavailable. Nevertheless, this gives us two potential sources of information. First, we can use the eye direction to estimate **oculomotor events** such as fixations and saccades [19]. Second, we can cast the ray into the 3D scene to know which objects the user was looking at. This is referred to as the user's **area of interest** (AOI) [19]. We identify four areas of interest based on our study design and task.

**Task**. This represents the orange button that the user must tap with their tool.

**Tool**. This represents the 3D geometry of the tool. For the 3p-hammer condition, this also includes the body of the avatar.

**Ambiguous**. This occurs when the ray intersects both task and tool.

**Elsewhere**. This represents all other elements of the scene (sometimes referred to as "whitespace" in the eye tracking literature [19]).

We refer to a distribution of AOI frames over a period of time as an **AOI allocation**. This allocation can be measured over any timescale, and it represents the proportion or percentage of a given AOI type to the size of the period.

A stream of AOI label events is also provided alongside the oculomotor events. When analyzing data, the first button press of each condition was thrown out due to the inability of a user to be in a tool extended state before the condition began.

*3.3.1 Subjective Measures.* We administered the Gonzalez-Franco and Peck Embodiment Questionnaire [13] to participants to record a subjective measure of their experiences in each condition. We asked a subset of the questions related to topics in the study from their body ownership, agency and motor control, and location of the body sections. Questions were asked on a 7 point Likert scale. We also collected an open-ended question asking "What strategies did you use to accomplish the task using this tool?" to deteremine if there were consistent approaches used for controlling each condition.

## 3.4 Apparatus

The study was performed on a Windows 10 laptop with an Nvidia RTX 3060M GPU. The study software was implemented in the Unity game engine. We used the Meta Quest Pro headset with the eye tracking feature enabled. VR was run in PCVR mode via the QuestLink feature on the headset.

## 3.5 Procedure

Users filled out a questionnaire regarding background information. We then showed each user a video recording depicting the headset view of someone performing the study. While the video played, we explained how the task works and the differences between each tool. Each user, regardless of their VR experience, was given a brief explanation of how the VR headset worked. We measured each user's interpupillary distance (IPD) with a phone app.

Users were then put into the VR headset, and the headset's eye tracking calibration program was run. In the user study software, a brief calibration process is performed to calculate user height and neutral hand positions.

In the study software, users first had a practice phase for each tool where they would practice the study task. Practice was performed in hammer → adaptation-hammer → 3p-hammer order. After users reported that they understood how the tool worked, up to a maximum of 30 button presses, they were moved on to the next tool, and finally into the first trial of the study.

Users were given tools in a randomized order, and each randomized order was run 2 times back to back, for a total of 60 button presses per tool.

After performing the task, users exited the VR headset and filled out a post-study questionnaire. On average, the complete session lasted about 30 minutes and users were compensated with a gift card for their time.

## 3.6 Participants

We recruited 18 participants from university mailing lists: 12 women, 4 men, and 2 non-binary. Participants included undergraduate students and graduate students. The mean age was 23 years old, ranging from 18 to 30 years. 6 participants had never used VR at all, 10 participants had used VR a few times, and 2 participants had used VR many times.

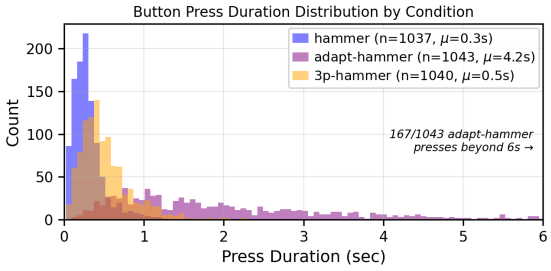



**Figure 3: Histogram showing the range of task completion times for each condition. The x-axis is cut off at 6s, trimming 167 button presses (16%) for the adaptation-hammer condition.**

## 3.7 Hypotheses

We have two hypotheses that we investigate with this experiment.

- **H1** - Eye tracking measures will show a difference between the hammer and adaptation-hammer conditions. This is based on prior understandings of tool extension, and our explicit design of these conditions.
- **H2** - The tool extension state, and thus the eye tracking patterns, for the 3p-hammer condition more closely resembles the hammer condition vs the adaptation-hammer condition. This is based on prior work in HCI investigating first vs third person body relationships [46].

# 4 Results

During the study, users recorded a total of 3132 button presses. 12 presses were dropped because user's positioned the tool in such a way that the button was immediately pressed when it appeared, for a total of 3120 valid presses.

We measure the mean and standard deviation of task completion time: **hammer** ($\mu = 0.30s, \sigma = 0.20s$), **adaptation-hammer** ($\mu = 4.17s, \sigma = 8.72s$), **3p-hammer** ($\mu = 0.51s, \sigma = 0.32s$). See Figure 3 for a histogram plot for each condition. Note the task completion times for the adaptation-hammer condition is spread out over a much larger range of values, indicating that this condition tested a much larger range of difficulties. These results show that the adaptation-hammer was indeed more difficult to control.

A Friedman test on participant-level mean press times showed a significant effect of condition ($Q = 36.000, p < 0.001$). Bonferroni-corrected Wilcoxon signed-rank tests found all three pairwise differences significant ($W = 0.0, p < 0.001$ in each case): every participant was slower with the 3p-hammer than the hammer, and slower again with the adaptation-hammer than the 3p-hammer.

Across all conditions, **task** gaze was the dominant AOI category, with means of 43.8% (hammer), 46.4% (adaptation-hammer), and 46.7% (3p-hammer). **Elsewhere** gaze was the second largest category, with means of 32.8%, 33.8%, and 29.9% respectively. **Tool** gaze was infrequent overall, with means of 6.7% (hammer), 8.0% (adaptation-hammer), and 8.1% (3p-hammer). **Ambiguous** gaze accounted for the remaining proportion across all conditions (means: 16.7%, 11.8%, 15.3%).

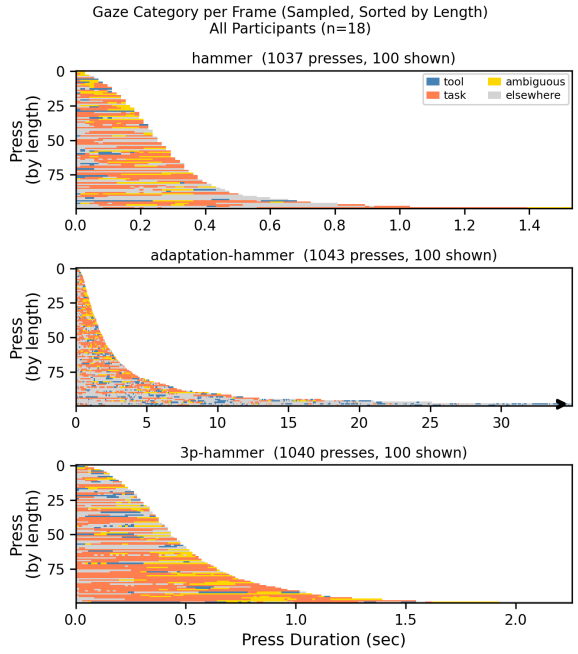



**Figure 4: Gaze category over time for a uniform sample of button presses in each condition, sorted by press duration (shortest at top). Each row represents one button press, with color indicating the gaze classification at each moment. Black arrows on the right edge indicate presses that extended beyond the 35-second display limit. Shorter conditions (hammer, 3p-hammer) show predominantly task-directed gaze, while adaptation-hammer reveals increased tool and elsewhere gaze in longer-duration presses.**

We provide a visual representation of AOI patterns via the scarf plots in Figure 4. Each row represents one button press in the given condition, and the width of the bar represents how long it took to press that button. Color indicates AOI markings at each point of time during the task. **Task** is orange, **tool** is blue, **ambiguous** is yellow, and **elsewhere** is grey. 80 uniformly sampled button presses are shown for each condition.

Visually, we can begin to see trends emerge. The hammer and 3p-hammer conditions appear to be mostly dominated by task gazes, with 3p-hammer having a possibly higher proportion of ambiguous gaze. The adaptation-hammer condition shows quite a bit more tool and elsewhere gaze for the more difficult trials.

To quantify the effect of each condition on AOI allocations, we perform a Friedman test comparing conditions for each AOI type (Figure 5). **Task** and **ambiguous** gaze showed no significant effect of condition (task: $Q = 3.577, p < 0.1$; ambiguous: $Q = 4.831, p < 0.1$), while **tool** ($Q = 14.550, p < 0.001$) and **elsewhere** ($Q = 8.333, p < 0.05$) did.

For AOI types with a significant omnibus effect, Bonferroni-corrected Wilcoxon signed-rank tests were run to identify pairwise differences. For **tool** gaze, adaptation-hammer was significantly higher than hammer ($W = 0.0, p < .05$), indicating that users looked at the tool more when the sensorimotor mapping was disrupted. For **elsewhere** gaze, adaptation-hammer was again significantly higher than hammer ($W = 27.0, p < .05$). No other pairwise comparisons reached significance.

One observation about the adaptation-hammer condition is that button presses spanned a wide range of task completion times. This means some configurations of adaptation-hammer parameters were easier than others. This is because small control-axis offsets were included in the parameterization and in these cases the adaptation-hammer performed similarly to a regular hammer. One benefit of this is that we can examine the single condition of adaptation-hammer to explore in more detail what makes a task "easy" vs "hard".

To do this, we split the adaptation-hammer tasks into upper and lower halves ($\eta = 1.9s$), as we posit that the lower half of tasks likely more resembles a tool extended state. Performing a Wilcoxon signed-rank test on each AOI category between these halves shows significant differences between **tool** values and **elsewhere** values, confirming the prior result. As a sanity check, we also perform the same test comparing the lowest (faster) half of adaptation-hammer to a uniform sampling of the same size from the hammer condition, and find no significant differences. This confirms the easiest adaptation-hammer presses resemble those from the hammer condition.

## 5 Predicting Tool Extension

Our results show that different eye patterns emerge based on different levels of tool extension, but this alone does not help us understand if a user is tool extended at a given point in time during an open-ended task. In open-ended use, we have no notion of when tasks begin or end, so we need a classifier that can work over windows of time.

We define the two primary tool extension states a user can be in: either **tool extended** or **tool breakdown** [52] (or extended vs breakdown, for short). In this section we describe building a random forest based classifier to predict tool extension vs breakdown state.

How can we tell from our user data if a user was in an extended or breakdown state? We can see that there are task completion time differences between the fast and slow button presses in the adaptation-hammer condition with regard to AOI allocation (see Figure 6). We can also clearly observe the visual differences between fast and slow button presses in Figure 4. From these two results we make an assumption: longer button press times in the adaptation-hammer condition must have exhibited tool breakdown states. We will use this fact for labeling purposes.

### 5.1 Labeling Strategy

For labeling, we use only the data from the adaptation-hammer condition because it is the only condition where users exhibited a breakdown behavior. We had no ground truth data about whether a user was in a tool breakdown state during a task, so our goal is to extract windows of time over the button presses in these conditions and auto-label them.

Our window sliding strategy has three variables. **Window size**, which is the total amount of time we collect for training and classification. If the window size is smaller than the task completion time, then the frames in the window are all labeled as a tool extended

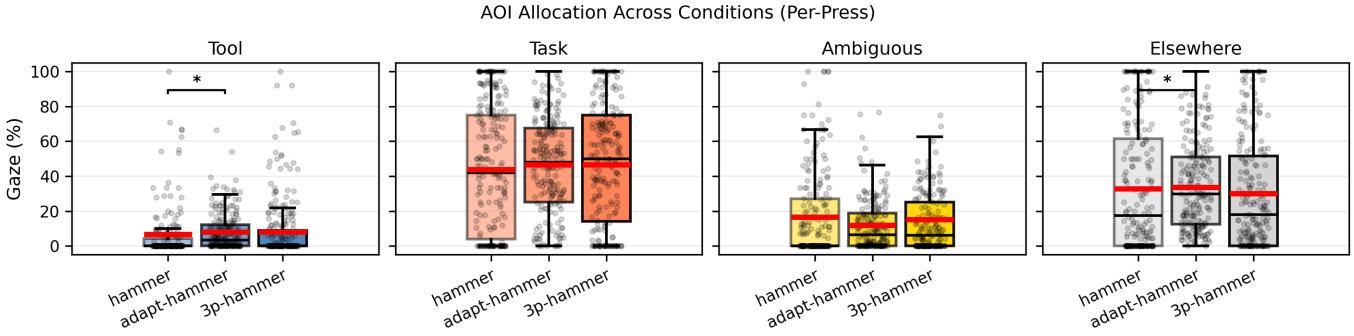



**Figure 5: Distribution of gaze allocation across four AOI categories for all button presses, grouped by condition. Red lines indicate means; black lines indicate medians. Significance brackets show post-hoc Wilcoxon signed-rank tests (Bonferroni corrected) following a significant Friedman omnibus test. Adaptation-hammer produces significantly more tool gaze and elsewhere gaze than hammer.**

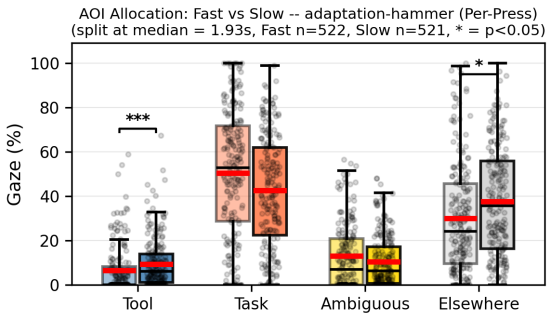



**Figure 6: Gaze allocation for fast and slow button presses in the adaptation-hammer condition, split at the median press duration ($\eta = 1.93s$). Red lines indicate means. Lighter boxes are fast presses; darker boxes are slow. Slow presses show higher tool and elsewhere gaze and lower task gaze, consistent with struggling users redirecting attention away from the task target.**

state. **Stride**, which is the distance we move a window forward when creating a new label. This determines how much overlap subsequent windows will have, which ends up having some effect on final classifier precision. **Look-ahead distance**, which is a discriminator that asks if the task will continue running for at least this amount of time. Each window is labeled based on whether the task continues for at least the look-ahead distance time beyond the window, providing a predictive signal that the user has not yet achieved task completion. This works because our experiment results showed that longer completion times correlate with an adjusted AOI allocation. The value of each of these parameters will be optimized during the classifier training process (see below).

## 5.2 Feature Extraction

To train the model, we need to know which eye gaze features are relevant to tasks where the user exhibited tool breakdown. We manually selected a number of features relevant from our experimental results, as well as eye tracking based research generally [19], and list them in Table 1 (See Table 2 in the Appendix for more detailed descriptions).

These features are computed from the individual frames (AOI classifications, fixations, saccades, etc) inside a given window. Each window produces a 20 dimensional vector: the computed value for each of the 19 features, and the extension vs breakdown classification from the look-ahead parameter.

One possible concern is that features related to run length could be biased for longer tasks (e.g. mean_run_length being correlated with total task length). To account for this, the sliding window is fixed-size, so all features are bounded by the same window regardless of where in the press they fall. A window at time = 2 and a window at time = 50 both contain exactly the same amount of data. Run lengths, counts, proportions — all are constrained to the same frame budget.

Longer tasks can produce more windows, which can lead to an imbalance in extension vs breakdown windows. During model training we saw 646 extension windows, and 9354 breakdown windows. To correct for this, we down-sample the breakdown class uniformly to match the same number of labeled windows as the extended class [16].

## 5.3 Classifier

We train a random forest classifier with 100 trees using the scikit-learn Python package. A random forest approach was chosen because it's a commonly used classifier in the eye tracking literature for this type of task [22, 45]. It handles mixed feature types (binary, proportional, continuous), provides built-in feature importance, and performs well with moderate sample sizes.

We utilize a hyperparameter tuning process via grid search [5] to optimize window size (1.0-3.0s), look-ahead distance (0.5s-2.0s), and decision threshold for the random forest classification (0.3-0.7), selecting a configuration that maximized F1 scores under the LOPO cross-validation.

**Table 1: Gaze features extracted from each 3-second sliding window. All features are rate-based or proportional, ensuring independence from window position or task duration. Full descriptions of each feature can be found in the Appendix.**

| Category | Features |
|---|---|
| AOI Proportion | `pct_tool`, `pct_task`, `pct_elsewhere`, `has_tool` |
| Gaze Transition | `switch_rate`, `transition_entropy` |
| AOI Run Length | `mean_run_length`, `std_run_length`, `max_run_length`, `mean_tool_run`, `n_tool_runs` |
| Eye Movement | `pct_fixation`, `pct_saccade`, `fixation_count`, `saccade_count` |
| Eye Dynamics | `mean_fixation_dur`, `mean_saccade_dur`, `mean_velocity`, `std_velocity` |

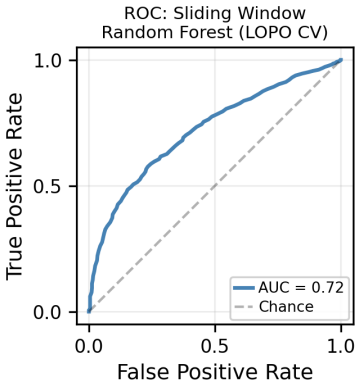



**Figure 7: ROC curve for the sliding-window Random Forest classifier evaluated with leave-one-participant-out cross-validation, showing the trade-off between true positive rate and false positive rate across decision thresholds (AUC = 0.72)**

## 5.4 Predictor Results

The highest performing model was selected to have a window size of 3.0s, a look-ahead value of 0.5s, and a decision threshold of 0.3. The trained model has an accuracy of 83.0%, precision of 94.9%, recall of 86.5%, and an F1 score of 90.5%. An ROC curve plot shows an AUC of 0.720, which is considered a moderate quality (See Figure 7). A confusion matrix plot of results can be seen in Figure 8, left.

To determine how well the model generalizes to user data it has not seen before, we run a LOPO cross-validation test. For each of the 18 participants, the model is trained on data from the other 17 and tested on the held-out participant. This repeats 18 times, so every participant serves as the test set exactly once. The reported metrics are aggregated across all 18 folds. The per-user results of this cross-validation are shown in Figure 8, right.

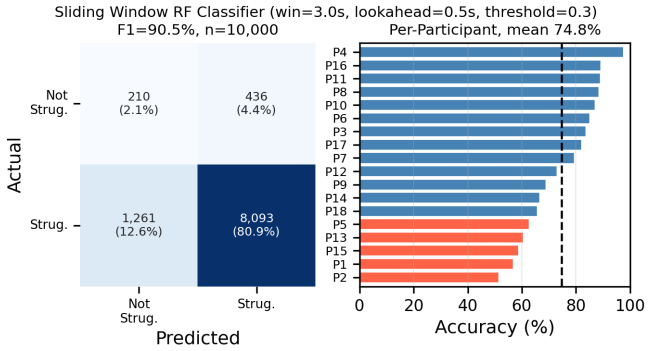



**Figure 8: (Left) Confusion matrix for the sliding-window Random Forest classifier aggregated across all LOPO folds. (Right) Per-participant classification accuracy under LOPO cross-validation; red bars indicate participants with accuracy below 65%.**

We also compute a feature importance metric, which is presented in Figure 9. We can see that the most relevant features to model success are AOI proportion related, such as pct_task and pct_elsewhere. Although ocular motion features such as saccades and fixations contribute, their effect is secondary.

## 6 Discussion

Eye tracking hardware is becoming a standard feature in VR headsets, making gaze a useful sensing channel for new software and interactions. This necessitates good strategies for operationalizing eye gaze data into software design features. One such direction is understanding the tool extension state of a user.

### 6.1 Condition Behaviors

A number of different behaviors can be observed from the study results.

**Hammer**. In the hammer condition, users completed button presses extremely fast. Their gaze was almost entirely task oriented, with very few glances elsewhere, and few glances toward the tool. This represents fluent, ready-to-hand tool use.

**Adaptation-Hammer**. In this condition, users had highly variable task completion times based on the difficulty of the necessary sensorimotor adaptation. Longer presses showed a characteristic behavior of early task glancing, and then longer periods of looking regularly toward the tool. We interpret the longer completion times as evidence with a present-at-hand state: the tool is no longer transparent, and the user must constantly monitor it.

**3p-Hammer**. Task completion times in this condition were slightly higher than in the hammer condition, but much faster and a tighter distribution than adaptation-hammer. Gaze patterns closely match hammer, being dominated by task and minimal tool. However, one difference is significantly more ambiguous frames, where the user's gaze looked at both task and tool. This could be explained by the perspective change: the AOI size of the task buttons is smaller, and the user must spend more time aligning the tool toward the button during the final phases of a task due to perspective shift.

## 6.2 Tool Extension Differences

With regards to **H1**: Does the user's gaze pattern change between our two baseline conditions: hammer (tool extended) and adaptation-hammer (not tool extended)? We observe two behavioral changes between these conditions. First, the adaptation-hammer runs had significantly longer and widespread completion times. Second, statistical tests show that users looked more at the tool and elsewhere during the adaptation-hammer condition. This result confirms prior work in the HCI literature suggesting higher ratios of tool attention during states where the user is not tool extended [3]. There is also prior work in Psychology suggesting a higher ratio of looking around the scene during unskilled action [12], however for our study, we observed users regularly glancing between the tool and task during tool breakdown with the adaptation-hammer, which could suggest that a higher proportion of elsewhere glances are present for different reasons (the "space" between the task and tool). Nevertheless, the signal is present in the study data.

We also use our study to gain new understandings about tool extension of VR-only interfaces. For **H2**: Does the 3p-hammer condition more closely resemble a tool extension or unextended state? The condition had statistically similar AOI distributions to the hammer condition, and although the task completion times were statistically higher than hammer, the distribution much more closely resembles hammer than adaptation-hammer. This confirms findings from prior work [46] suggesting third person bodies have low embodiment, but high agency ratings (which describe a tool-like relationship). Users in the 3p-hammer condition do achieve a state of tool extension, showing that tool extension can apply in situations with radically different (and uniquely VR) body configurations.

Completion time is a proxy in our data; we provide two converging lines of evidence that it tracks the ready-to-hand to present-at-hand states. First, subjective control ratings distinguish breakdown from slowness. Embodiment questionnaire ratings show that subjective control over the tool drops sharply from the hammer to the adaptation-hammer (5.83 to 2.44, $p = .0004$), alongside a roughly 14x increase in per-press completion time (0.30s to 4.17s). The 3p-hammer can serve as a control: it is also significantly slower than the hammer (0.51s vs 0.30s per press, $p < .0001$), yet participants felt control was preserved (5.56, n.s.). If slowness alone drove the loss of control, the slower third-person condition should show it too; it does not. This rests on agency and control being a reasonable index of the phenomenological tool state, which prior work supports [13]. Second, in open-ended questions users also described having to probe the mapping space before acting in adaptation-hammer, which indicates tool breakdown, not just difficulty. This suggests that completion time within the condition reflects how quickly each user resolved the initial breakdown: slow adaptation-hammer trials are indicative of sustained breakdown states. These are post-hoc validations on a fixed dataset; future work could collect per-trial subjective probes more directly.

## 6.3 Tool Extension Classifier

Looking to the effect of feature importance on the tool extension classifier (Figure 9), we can see that the percentage of user gaze directed towards the task or elsewhere were the most important

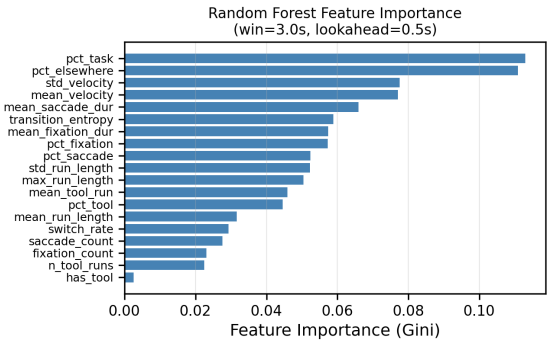



**Figure 9: Mean decrease in impurity across LOPO folds for each gaze feature, showing that AOI proportion features (pct_tool, pct_task) are the strongest predictors of struggling.**

features. Oculomotor dynamics features such as percentages of fixations and saccades were less important. This suggests *what* a user looks at is more important than *how* a user looks, with regard to understanding their tool extension state. A practical implication of this is that even a simple AOI based classification system might be able to capture most of the necessary signal for real-time detection.

One observed difference between the experiment and the classifier results is that pct_task (presence of task glances) was the most important feature when discriminating between tool extension and breakdown, but showed no significant difference between "slow" and "fast" button presses for adaptation-hammer (see Figure 6). One explanation could be due to task glances accumulating over long button press times, but less in the shorter 3s window of the classifier. Thus, their presence during a window is more important than their aggregate presence over a full, difficult button press.

The LOPO evaluation of the classifier shows that this generalizes across participants, but with some notable variance. Classifier accuracy ranged from 51.5-97.5% across users. The low accuracy participants show a distinct pattern: they exhibit mostly task oriented glances, rarely looking at the tool during the adaptation-hammer condition, and they generally exhibited lower completion times. These participants may be using a different oculomotor behavior strategy of keeping their eyes on the task, using peripheral vision to control the tool. This represents a boundary condition for the classifier, which can only identify participants that are experiencing tool breakdown through their visual eye movement patterns. We include a scarf plot of 10 runs from every participant in the study in the Appendix (see Figure 11), which visually demonstrates this difference in behavior.

# 7 Design Space

Our results show it's possible to construct a real-time metric that can accurately predict whether or not users are in a tool extended state (*Section 5.4*). This classifier enables future systems to enact timely interventions across various applications.

Through literature review, observation of user behavior during our study, and our own design ideation, we identify a design space for using tool extension prediction to develop novel *attentive* and

*adaptive* systems. It is comprised of three dimensions: **Goal Type**, **Intervention Direction**, and **Flow Disruption**. The combination of these dimensions span a wide space of potential tool extension driven intervention designs. See Figure 10.

*7.0.1 Goal Type.* When determining the Goal Type, the primary question the designer must ask is whether the primary purpose of the task is to **train**, **entertain** or **observe**. If the primary goal is *training*, our metric provides insight into whether the user is struggling or succeeding, allowing the system designer to create timely educational interventions. If the goal is *entertaining*, this metric can provide a measure of when a task is beginning to become too difficult or easy rather than enjoyable. If the goal is *observation*, this metric can be used to infer a user's level of tool extension as an indicator of their familiarity and interaction patterns with a given tool, where higher levels of tool extension may reflect more efficient, goal-directed behavior. Goal Type is the most design-driven of the three dimensions; it does not follow from a specific result, but enumerates the applications our predictor enables.

*7.0.2 Intervention Direction.* Once given information regarding the user's state, the designer must decide whether they would like to push the user *towards tool extension*, *away from tool extension*, or make *no change at all*. Using this metric, the designer can create timely interventions that **support**, **maintain**, or **challenge** the user's current level of tool extension according to the ultimate goal of the interaction. This dimension follows from our conditions showing that task difficulty moves the tool extension state in a measurable way.

*7.0.3 Flow Disruption.* Our proposed metric provides information about the user's level of tool extension *while* they are performing a tool-assisted task. Thus, the system designer must decide how disruptive an intervention based on the metric should be to the user's current process. The highest degree of disruption would be to **inform**, introducing a **new element or view** to the scene to directly provide information to the user, moving attention away from the task. An intermediate degree of disruption would be to **demo**, or augment *existing* elements embedded in the task to demonstrate a particular movement or action. Finally, a minimal degree of disruption would be to **adjust**, which would include silently altering the mechanics of the scene. Flow Disruption follows from our gaze results: extended users keep their gaze on the task while struggling users already look around, so the prediction metric shows when an interruption is costly versus free.

## 7.1 Applications

We describe a collection of examples enabled by our classifier by sampling across the design space. For each example, we provide a description, its position within the design space, and describe its relationship to preexisting research directions [42].

*7.1.1 Fantasy Battle (Entertain, Challenge, Adjust).* Consider a simple VR fantasy game in which a user must battle multiple enemies in a stadium. The user is holding a weapon, such as a sword or axe, and inflicts damage on the opponents by contacting them (*task*) with the weapon (*tool*). A new form of upgrade path can be made available. As users become proficient with weapon use, newer, more powerful weapons can be unlocked that are also more difficult to control from a tool extension perspective, affording the player new challenges as soon as they master old ones. Unlike performance-based methods of determining task difficulty, this method allows adjustments to be made more proactively, based on the user's eye-tracking data over the course of a few seconds, rather than reactively, after performance outcomes have already signaled a mismatch in challenge level. This **adjustment** increases the difficulty of the task, thus **challenging** the user, preempting a state of boredom, and increasing **entertainment** value.

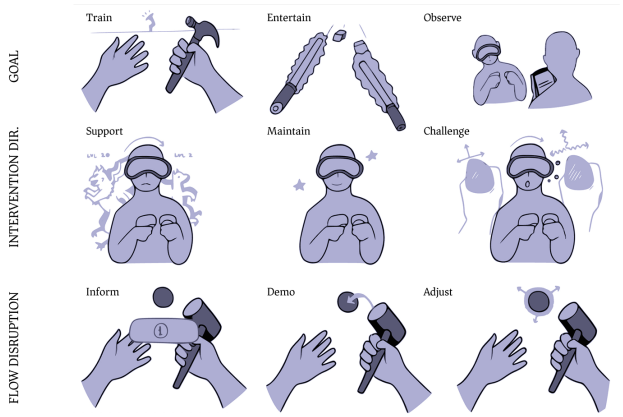



**Figure 10: Design space enabled by the proposed real-time tool extension metric, including Goal, Intervention, Direction, and Flow Disruption.**

This example demonstrates how our proposed metric can extend prior work leveraging **eye-tracking data for activity recognition.** In this game, the degree of tool extension can be used as a proxy for engagement with a tool-assisted activity. Using this measure, the Fantasy Battle is able to develop an attentive interface that is able to make game-adjustments to maintain user engagement.

*7.1.2 Furniture Assembly Assistant (Train, Support, Demo).* When augmented with computer-vision scene labeling to determine AOI, a real-time measure of tool extension can also be beneficial in Augmented Reality (AR). Consider an AR application with the goal of helping a user assemble a specific furniture set (*task*). The assembly kit features a collection of several distinct tools and pieces (*tool*). As the user assembles the furniture, our metric indicates to the system that the user is having difficulty using a specific tool, as measured by low tool extension. At this time, a demonstrative AR overlay is placed over the existing tool, showing the user how to properly use the given tool to install the appropriate screw. Instead of waiting for a certain amount of idling to indicate confusion, this method allows the system to preempt frustration in real-time. In this way, the metric allows for a timely **training** intervention that **supported** the user's goal by **demonstrating** proper tool use.

This example demonstrates how our proposed metric can extend prior work leveraging **eye-tracking data for visual presentation.** In Furniture Assembly Assistant, the tool extension metric is a proxy for user confusion - if they are having difficulty using the given tool, it should be indicated by a lack of tool extension. Once this confusion is detected, the visual presentation of the scene

changes, showing a demo of proper tool use. In this way, Furniture Assembly Assistant is an adaptive interface that is more robust to unpredictable instances of user confusion.

*7.1.3 Interaction Evaluation Tool (Observe, Maintain, Inform).* Not all uses of these metrics must encompass real-time interventions. Consider a system designed to support designers in evaluating the usability of a VR application prototype. In this application, users are asked to complete a series of tasks (*task*) using the system's controls or mechanics (*tool*), while our proposed metric is used to **observe** and infer their level of tool extension throughout the interaction. Lower levels of tool extension may indicate moments of uncertainty, confusion, or inefficiency, while higher levels may reflect more fluent, goal-directed interaction. In this application, the system **maintains** the task conditions without introducing interventions, ensuring that user behavior remains unbiased. Instead of adapting the experience in real time, the metric is used to aggregate observations across users and sessions. These patterns are summarized in a report that can **inform** designers regarding user tool extension. Designers can then analyze this data to identify regions of an interface or level that consistently produce lower tool extension, suggesting potential areas of difficulty or poor usability.

This enables the metric to function as a proxy for interaction quality, informing iterative design processes such as level balancing in games or interface refinement in interactive systems. This example demonstrates how our proposed metric can extend prior work leveraging **eye-tracking data for usability analysis and design evaluation,** providing a continuous, behavior-driven signal of user experience without requiring explicit feedback from users.

## 8 Limitations & Future Work

One limitation of our current approach is the assumption that task completion time is correlated with tool breakdown, which is our labeling strategy for our classifier. As we can see from the variable user behavior (see Figure 11 in the Appendix), this assumption may not hold true. It's difficult to get an individualized task difficulty/tool extension state rate from users when the explicit goal of the study is to measure tool extension states, because any interruptions to probe the user about this may interrupt their tool extension state. Future work could design a study to try and infer this through other means, such as through crossmodally congruent and incongruent stimuli.

Our work is a *proof-of-concept* that a classifier can be built. Future work on generalization should look beyond single task with single tool morphologies. VR interaction is built from a small set of atomic tasks, such as target acquisition and selection, steering or trajectory-following, and gesture [7, 11], each with different motor and visual demands. Our task is an acquisition-style task, and it is unknown whether the gaze signatures that discriminate tool extension here transfer to tasks with different motor demands, such as continuous manipulation, bimanual coordination, or precision assembly. One path toward general deployment is to train a separate model for each interaction primitive. If an application's required primitives are known in advance, an ensemble that routes between the relevant per-primitive models could give coverage across applications without retraining for each one.

Per-user reliability ranges widely (51.5-97.5%, mean 74.8%), and some of this comes from users who rarely look at the tool during breakdown. A short calibration phase could predict classifier performance for individuals. By walking the user through a fluent baseline tool and a disrupted one, we collect labeled gaze for that individual and can estimate their personal classifier accuracy directly, rather than inferring it from completion time. An application can then set a target accuracy and enable the tool extension feature only for users who clear it. At a 65% target, for example, around 5 of our 18 participants would be flagged as unreliable. Compensating for those users is harder. With only a handful of them in our data, we can't confidently characterize what their vision strategies look like. If these strategies turn out to be shared, or drawn from a small set, one option is to train one model per strategy and route each user to the right one after calibration. We leave this to future work, as it would need a larger, strategy-diverse sample.

Another limitation is that our study only uses the single eye ray available on the Meta Quest Pro headset. Additional binocular features such as vergence depth may carry additional signal about whether a user is focusing at different depths, such as on the tool plane or task plane.

Future work should also include more longitudinal studies examining how gaze patterns shift as users learn a tool over several sessions rather than minutes. Does the classifier track the gradual transition from present-at-hand to ready-to-hand over days of practice? Do users ultimately converge on gaze patterns, or do all users uniquely have their own gaze pattern signatures?

## 9 Conclusion

In this paper, we studied gaze behavior during tool use in VR across three conditions spanning a range of tool extension states. Our results show that where users look, specifically whether they visually check the tool, reliably distinguishes tool extension from tool breakdown. This maps onto the phenomenological distinction between ready-to-hand (tool is transparent) and present-at-hand (user consciously monitors the tool). We developed a sliding window Random Forest classifier that predicts if the user is not tool extended in real time, reaching a 90.5% F1 with 94.9% precision during a LOPO evaluation. This work demonstrates that standard VR headset eye trackers can serve as a practical, real-time measure of tool extension state. The design implications of this metric are discussed with a design space analysis, and several example applications enabled by our results.

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# A Appendix

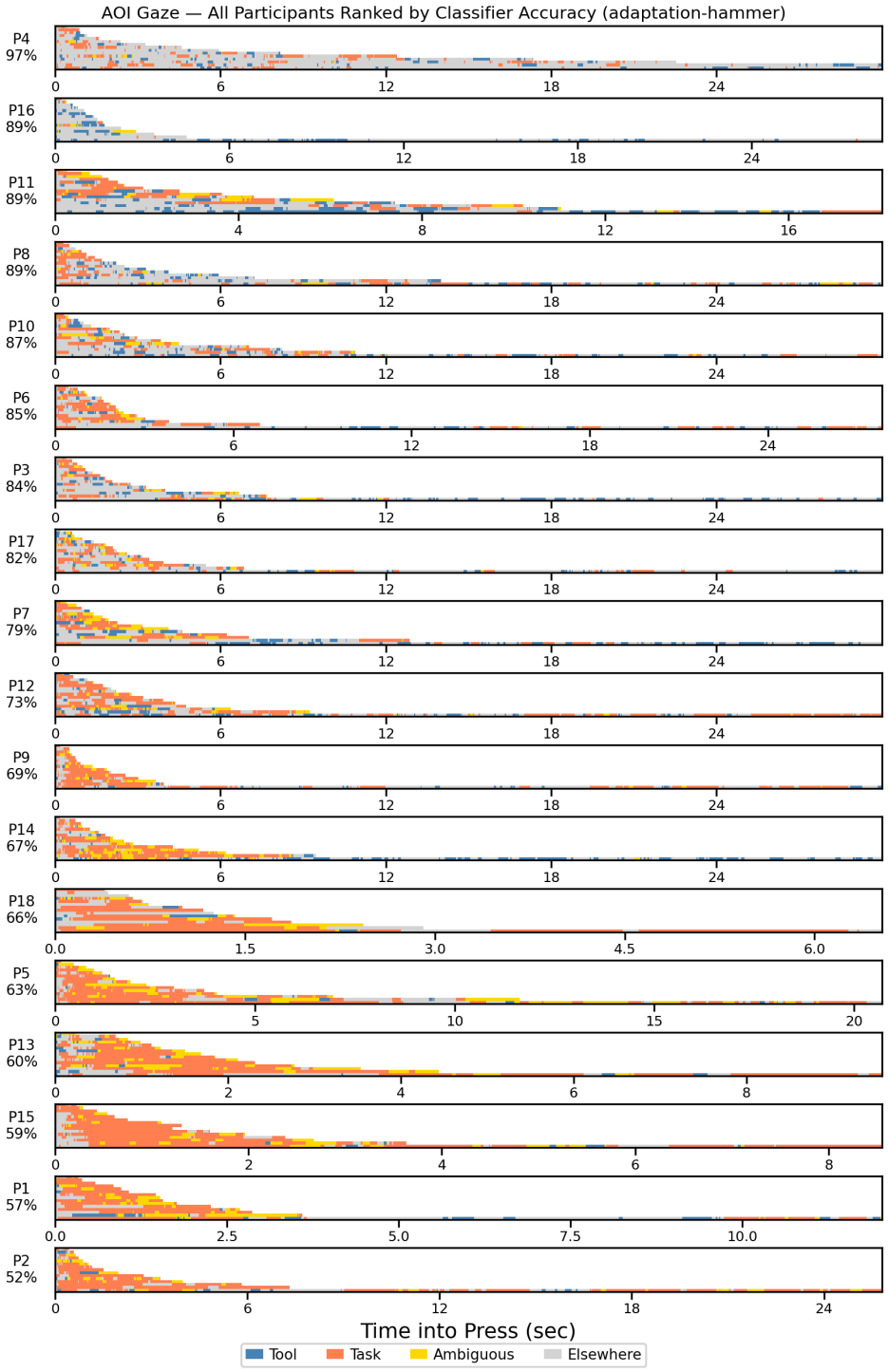



**Figure 11: Gaze category over time for a uniform sample of each user for the adaptation-hammer condition, ordered by the performance of the classifier on their data. A clear trend can be seen where AOI allocations for users where the classifier worked well are dominated by tool and elsewhere gaze.**

**Table 2: Gaze features extracted from each 3-second sliding window. All features are rate-based or proportional, ensuring independence from window position or task duration.**

| Category | Feature | Description |
|---|---|---|
| AOI Proportion | `pct_tool` | Proportion of frames on tool AOI |
| | `pct_task` | Proportion of frames on task AOI |
| | `pct_elsewhere` | Proportion of frames elsewhere |
| | `has_tool` | Binary: any tool gaze in window |
| Gaze Transition | `switch_rate` | AOI transitions per frame |
| | `transition_entropy` | Entropy of AOI transition matrix |
| AOI Run Length | `mean_run_length` | Mean consecutive frames in same AOI |
| | `std_run_length` | Std. dev. of run lengths |
| | `max_run_length` | Longest uninterrupted AOI dwell |
| | `mean_tool_run` | Mean duration of tool gaze episodes |
| | `n_tool_runs` | Number of tool gaze episodes |
| Eye Movement | `pct_fixation` | Proportion of fixation frames |
| | `pct_saccade` | Proportion of saccade frames |
| | `fixation_count` | Number of fixation episodes |
| | `saccade_count` | Number of saccade episodes |
| Eye Dynamics | `mean_fixation_dur` | Mean fixation duration (normalized) |
| | `mean_saccade_dur` | Mean saccade duration (normalized) |
| | `mean_velocity` | Mean angular velocity (°/s) |
| | `std_velocity` | Std. dev. of angular velocity |